Geopolitical AI-Powered Stock Intelligence & Market Impact Analysis Platform

Author: MarketAtlas Quantitative Research Team

Version: 2.0 (Expanded Foundation Model Edition)

1. Project Objective

The objective of this platform is to develop a deployable, multi-agent AI system that analyzes geopolitical events, global news, and financial market data to provide explainable stock recommendations and risk assessments. Unlike traditional time-series models that rely solely on historical price feeds, this system incorporates real-time geopolitical intelligence, semantic consensus deduplication, and topological graph-theory causal mapping to capture market shocks.

2. Core System Pipeline

The platform coordinates seven highly specialized AI agents in a state-graph workflow (inspired by LangGraph):

1. **News Agent:** Ingests global news streams (GDELT/ACLED), cleans titles, and performs TF-IDF Cosine semantic clustering to deduplicate syndications.
2. **Event Intelligence Agent:** Conducts NLP sentiment classification (FinBERT/DeBERTa) and entity extraction (spaCy NER) to parse structured event metadata.
3. **Market Data Agent:** Monitors price momentum, 14-day rolling volatility, and raw volume status.
4. **Knowledge Graph Agent:** Maps relationships topologically in-memory (NetworkX) and computes causal risk propagation weights.
5. **Impact Analysis Agent:** Synthesizes direct severity and graph path propagation to output a composite risk index [0.0, 1.0].
6. **Recommendation Agent:** Formulates BUY/HOLD/SELL signals using domain-aware hedging heuristics.
7. **AI Report Generation Agent:** Compiles all intermediate traces into a publication-grade markdown investment thesis.

3. Foundation Model & Advanced Relational Upgrades

To elevate this platform to a 'Foundation Model Class' system suitable for institutional quantitative trading, we have integrated the following critical relational and algorithmic enhancements:

A. Semantic GDELT Consensus Deduplication: GDELT is noisy. The Consensus Engine uses TfidfVectorizer and cosine clustering (HDBSCAN prototype) to group hundreds of redundant syndications into a single discrete Super Event cluster, tracking media momentum (cluster size) and source diversity to eliminate duplicate frequency bias.

B. Spatio-Temporal Graph Propagation: Geopolitical risk is network-driven. The Knowledge Graph Agent projects topological dependencies: *[Event] -> [Country] -> [Component] -> [Sector] -> [Asset]*. Risk is propagated along the network path and dampened at a factor of 0.85 per step, allowing the system to flag vulnerabilities even when the stock is not directly mentioned in the headlines.

C. Double Machine Learning (DML) Causal Severity: Rather than using subjective severity scores, the Severity Scorer contains a causal calibration engine that regresses historical category occurrences against historical abnormal sector returns, estimating unbiased causal treatment coefficients ($ heta$) to weight events.

D. Bayesian Black-Litterman Portfolio Mapping: The recommendation logic is designed to be fed directly into the Black-Litterman model, blending general market equilibrium weights with the expected returns and confidence intervals formulated by the AI agents.

4. Implementation & Execution Status

The complete multi-agent pipeline is **fully functioning and implemented** inside the marketatlas/ package in the workspace. It includes native implementations of all 7 agents, the State Graph Orchestrator, and the complete TF-IDF consensus clustering, DeBERTa/FinBERT sentiment scoring, and NetworkX topological shortest-path graph calculations. The CLI runner run_platform.py successfully executes the pipeline on target assets (e.g., NVIDIA) and outputs serializable JSON traces.